

GORDON GOUGER

ggouger.com

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EXPERIENCE

Senior Machine Learning Engineer, Ozni AI

September 2025 – Present

- Machine learning lead for two SBIR programs totaling \$4M in funding.
- Responsible for intelligent sensing and Pattern of Life (PoL) analytics across vision, audio, and RF modalities.
- Developed and deployed drone-based geolocation algorithms leveraging onboard sensing and edge AI inference.
- Designed and implemented end-to-end ML systems spanning data ingestion, model training, deployment, and field testing.
- Deployed optimized models to constrained hardware including Raspberry Pi-class devices using ONNX and TensorFlow Lite.
- Owned full software and hardware architecture including sensor integration, edge compute selection, and power constraints.
- Built and maintained fully self-hosted CI/CD infrastructure using GitLab for model training, validation, and deployment.
- Curated and managed multi-terabyte datasets for signal detection, classification, and model validation using real sensors and simulators.

Principal AI Software Engineer, Northrop Grumman

October 2021 – September 2025

- Machine learning technical lead on multiple rapid prototyping and transition-to-production programs supporting operational systems.
- Developed CNN-based modulation recognition models for narrowband and wideband RF, including few-shot Siamese architectures.
- Deployed optimized modulation recognition pipelines combining classical DSP features (e.g., kurtosis) with CNN inference for IARPA programs.
- Designed and trained transformer-based models for overhead imagery, multi-spectral, hyperspectral, and wideband spectrum monitoring.
- Built and deployed end-to-end emitter geolocation algorithms using Hawkeye 360 data, including intercept-to-link-set generation.
- Performed intelligence fusion across RF, vision, and metadata sources for emitter classification and localization.
- Deployed ML models to FPGA, NVIDIA Orin, and embedded edge platforms using ONNX and TensorRT.
- Created and curated multi-terabyte RF datasets using real radios and high-fidelity simulators.
- Built air-gapped systems utilizing locally hosted LLMs for analysis, reasoning, and operator support.
- Led and mentored multiple ML engineers; served as ML lead on programs with cross-discipline teams.
- Regularly interacted with customer leadership and supported deployment into live operational environments.

Engineer, Rincon Research Corporation

June 2021 – October 2021

- Developed end to end GPS simulator.

EDUCATION

Master of Science in Electrical Engineering

May 2021

GPA: 4.00

Bachelor of Science in Engineering Physics

May 2020

GPA: 3.96

TECHNICAL STRENGTHS

FAA 107 Certified, Python, C++, C, MATLAB, Rust, CNNs, Transformers, Classical DSP + ML Hybrids, Modulation Recognition, Geolocation, Sensor Fusion, ONNX, TensorRT, TensorFlow Lite, Docker, Linux, GitLab CI/CD, Azure, Agile, Jira, $\text{\LaTeX}$

CLEARANCE

Active (but not currently held) TS/SCI

HOBBIES & INTERESTS

Running, Woodworking, Martial Arts, Glassblowing, Active in Local Church